

2008 Ford Edge SE

2008 ENGINE PERFORMANCE Electronic Engine Controls - Edge & MKX

2008 ENGINE PERFORMANCE

Electronic Engine Controls - Edge & MKX

SPECIFICATIONS

Material

Item	Specification	Fill Capacity
High Temperature Nickel Anti-Seize Lubricant XL-2 (US); CXG-2-B (Canada)	ESE-M12A4-A	-
Motorcraft Premium Gold Engine Coolant with Bittering Agent (US only) VC-7-B (US); CVC-7-A (Canada); or equivalent (yellow color)	WSS-M97B51-A1	-
Motorcraft SAE 5W-20 Premium Synthetic Blend Motor Oil XO-5W20-QSP (US); Motorcraft SAE 5W-20 Super Premium Motor Oil CXO-5W20-LSP12 (Canada); or equivalent	WSS-M2C930-A	-
Penetrating and Lock Lubricant XL-1 (US); CXC-51-A (Canada)	-	-

TORQUE SPECIFICATIONS

Description	Nm	lb-ft	lb-in
Camshaft position (CMP) sensor bolt	10	-	89
Catalyst monitor sensor	48	35	-
Crankshaft position (CKP) sensor bolt	10	-	89
Cylinder head temperature (CHT) sensor	10	-	89
Heat shield bolt and nut	10	-	89
Heated oxygen sensor (HO2S)	48	35	-
Knock sensor (KS) bolt	20	15	-
Mass airflow (MAF) sensor screws	2	-	18
PCM nuts	5	-	44
Variable camshaft timing (VCT) oil control solenoid bolt	10	-	89

DESCRIPTION AND OPERATION

ELECTRONIC ENGINE CONTROLS

The electronic engine controls consist of the:

- PCM.
- cylinder head temperature (CHT) sensor.

2008 Ford Edge SE

2008 ENGINE PERFORMANCE Electronic Engine Controls - Edge & MKX

- camshaft position (CMP) sensor.
- crankshaft position (CKP) sensor.
- mass air flow (MAF) sensor.
- heated oxygen sensor (HO2S).
- catalyst monitor sensor.
- variable camshaft timing (VCT) oil control solenoid.
- knock sensor (KS).

The PCM carries out the following functions:

- accepts input from various engine sensors to compute the fuel flow rate necessary to maintain a prescribed air/fuel ratio throughout the entire engine operational range.
- outputs a command to the fuel injectors to meter the appropriate quantity of fuel.

The CHT sensor:

- sends the PCM a signal indicating cylinder head temperature.
- resistance decreases as cylinder head temperature increases.

The CMP sensor:

- sends the PCM a signal indicating camshaft position used for fuel synchronization.

The CKP sensor:

- sends the PCM a signal indicating crankshaft position.
- is essential for calculating spark timing.

The MAF sensor:

- uses a hot wire sensing element to measure the amount of air entering the engine. Air passing over the hot wire causes it to cool.

The HO2S:

- creates a voltage signal dependent on exhaust oxygen content.
- provides feedback information to the PCM used to calculate fuel delivery.

The catalyst monitor sensor:

- monitors oxygen content after it flows through the catalytic converter.
- provides a voltage to the PCM used to calculate catalytic converter integrity.

The KS:

- is used to detect engine detonation.
- sends a voltage signal to the PCM.
- is able to provide a signal which retards the ignition timing, as necessary.

The VCT oil control solenoid:

- is an electrically controlled hydraulic valve that directs engine oil to the camshaft phaser. Once the PCM transmits a signal, the solenoid moves a valve spool, directing oil into the camshaft phaser cavity. This action changes valve timing by either inducing an advance or retard condition. The camshaft is, thereby repositioned in relation to crankshaft timing and allows for optimum engine performance and lower emissions.

DIAGNOSTIC TESTS

ELECTRONIC ENGINE CONTROLS

Refer to the **Introduction - Gasoline Engines** article.

REMOVAL AND INSTALLATION

CAMSHAFT POSITION (CMP) SENSOR

Material

Item	Specification
Motorcraft SAE 5W-20 Premium Synthetic Blend Motor Oil XO-5W20-QSP (US); Motorcraft SAE 5W-20 Super Premium Motor Oil CXO-5W20-LSP12 (Canada); or equivalent	WSS-M2C930-A

2008 Ford Edge SE

2008 ENGINE PERFORMANCE Electronic Engine Controls - Edge & MKX

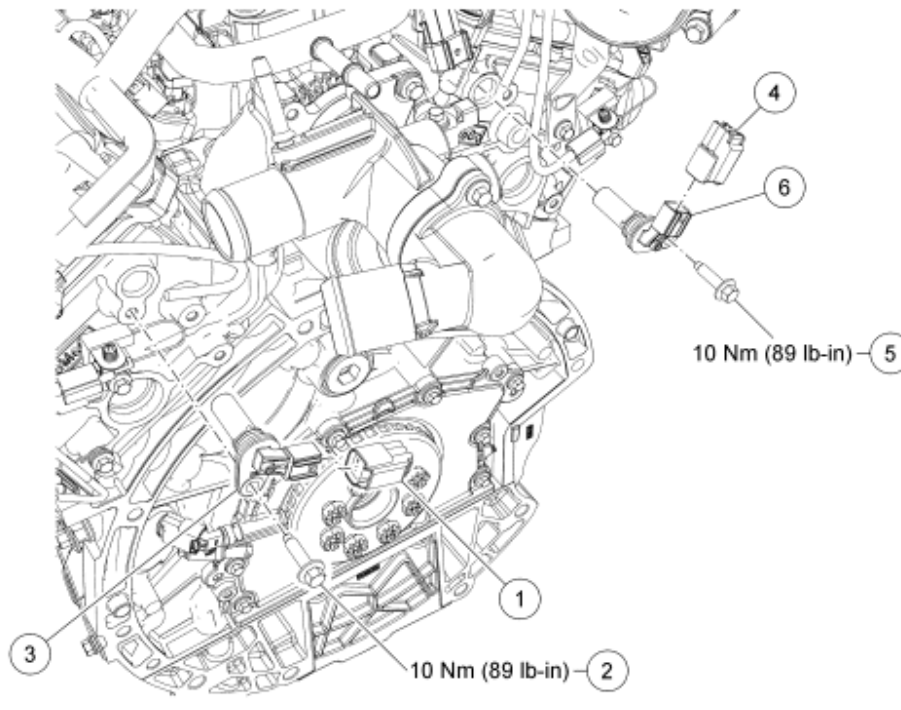


Fig. 1: Exploded View Of Camshaft Position (CMP) Sensor With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	14A464	LH camshaft position sensor (CMP) electrical connector
2	W500214	LH CMP bolt
3	6B288	LH CMP
4	14A464	RH CMP electrical connector
5	W500214	RH CMP bolt
6	6B288	RH CMP

REMOVAL AND INSTALLATION

RH sensor

1. Remove the air cleaner outlet pipe. For additional information, refer to **INTAKE AIR DISTRIBUTION AND FILTERING** article.

LH sensor

2. Remove the air cleaner assembly. For additional information, refer to **INTAKE AIR DISTRIBUTION AND FILTERING** article.

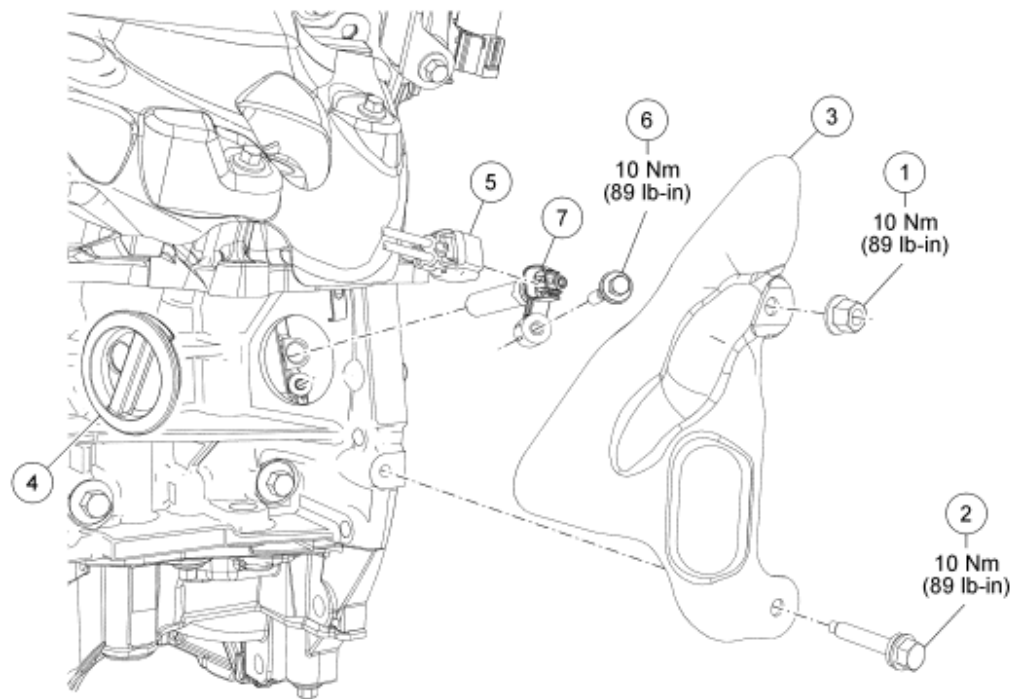
Both sensors

3. Disconnect the camshaft position sensor (CMP) electrical connector.
4. Remove the bolt and the CMP sensor.
 - To install, tighten to 10 Nm (89 lb-in).

NOTE: Lubricate the CMP O-ring seal with clean engine oil.

5. To install, reverse the removal procedure.

CRANKSHAFT POSITION (CKP) SENSOR



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Fig. 2: Exploded View Of Crankshaft Position (CKP) Sensor With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W701542	Heat shield nut
2	W701669	Heat shield bolt
3	6K342	Heat shield
4	6C070	Rubber grommet cover
5	14A464	Crankshaft position (CKP) sensor electrical connector
6	6D327	CKP sensor bolt
7	6C315	CKP sensor

REMOVAL AND INSTALLATION

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.
2. Remove the LH catalytic converter. For additional information, refer to **EXHAUST SYSTEM** article.
3. Remove the bolt, nut and the heat shield.
 - To install, tighten to 10 Nm (89 lb-in).
4. Remove the rubber grommet cover.
5. Disconnect the crankshaft position (CKP) sensor electrical connector.
6. Remove the bolt and the CKP sensor.
 - To install, tighten to 10 Nm (89 lb-in).
7. To install, reverse the removal procedure.

POWERTRAIN CONTROL MODULE (PCM)

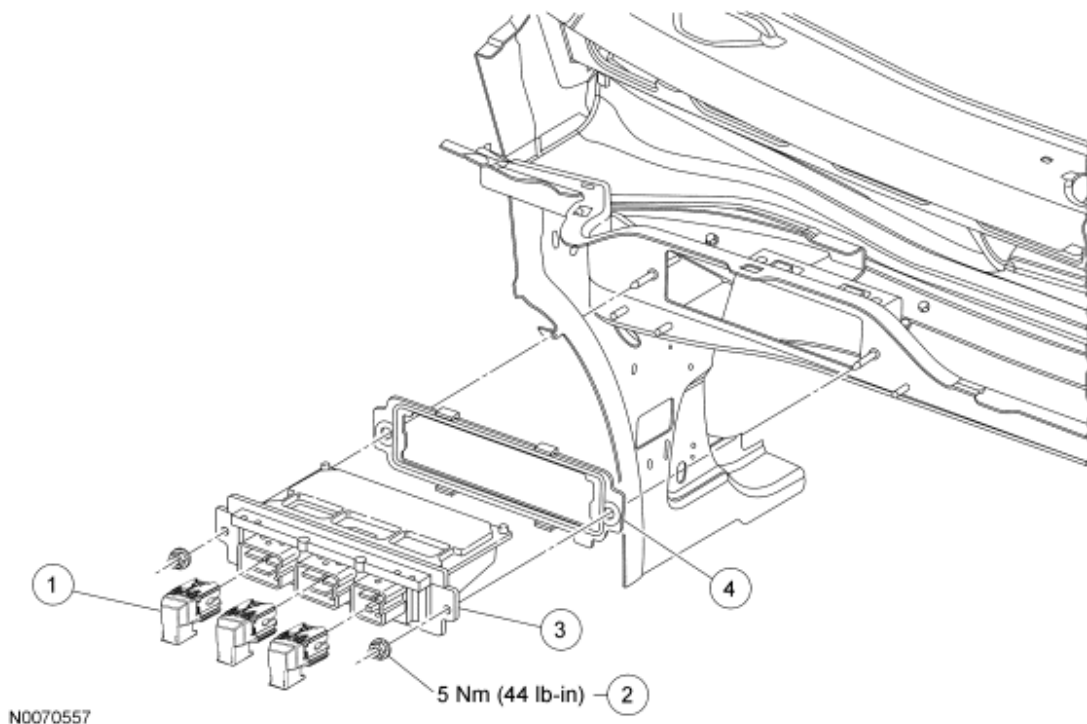


Fig. 3: Exploded View Of Powertrain Control Module (PCM) With Torque Specification
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	14A464	PCM electrical connector (part of 12B637) (3 required)
2	N804795	PCM nut (2 required)
3	12A650	PCM
4	12A696	PCM gasket

REMOVAL

NOTE: PCM replacement **DOES NOT** require new keys or programming of keys.

1. Retrieve the module configuration. Carry out the module configuration retrieval steps of the Programmable Module Installation procedure. For additional information, refer to **MODULE CONFIGURATION** article.
2. Disconnect the 3 PCM electrical connectors.
3. Remove the 2 nuts and the PCM.
 - Remove the gasket.

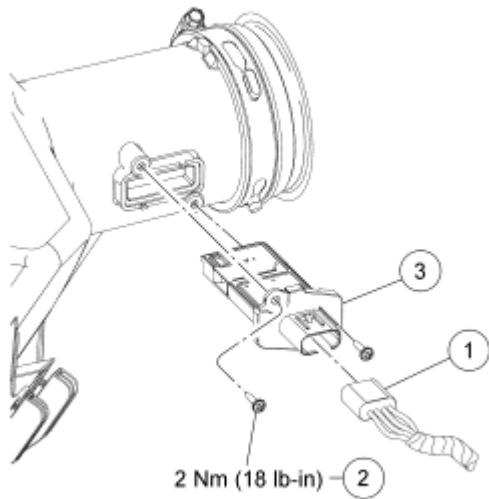
INSTALLATION

1. Install the gasket and the PCM.
2. Install the 2 PCM nuts.
 - Tighten to 5 Nm (44 lb-in).
3. Connect the 3 PCM electrical connectors.
4. Restore the module configuration. Carry out the module configuration restore steps of the Programmable Module Installation procedure. For additional information, refer to **MODULE CONFIGURATION** article.
5. Reprogram the passive anti-theft system (PATS). Carry out the Parameter Reset procedure. For additional information, refer to **ANTI-THEFT - PASSIVE ANTI-THEFT SYSTEM (PATS)** article.

MASS AIR FLOW (MAF) SENSOR

2008 Ford Edge SE

2008 ENGINE PERFORMANCE Electronic Engine Controls - Edge & MKX



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Fig. 4: Exploded View Of Mass Air Flow (MAF) Sensor With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	14A464	Mass air flow (MAF) sensor electrical connector
2	W709287	MAF sensor screw (2 required)
3	12B579	MAF sensor

REMOVAL AND INSTALLATION

1. Disconnect the mass air flow (MAF) sensor electrical connector.
2. Remove the 2 screws and the MAF sensor.
 - To install, tighten to 2 Nm (18 lb-in).
3. To install, reverse the removal procedure.

CYLINDER HEAD TEMPERATURE (CHT) SENSOR

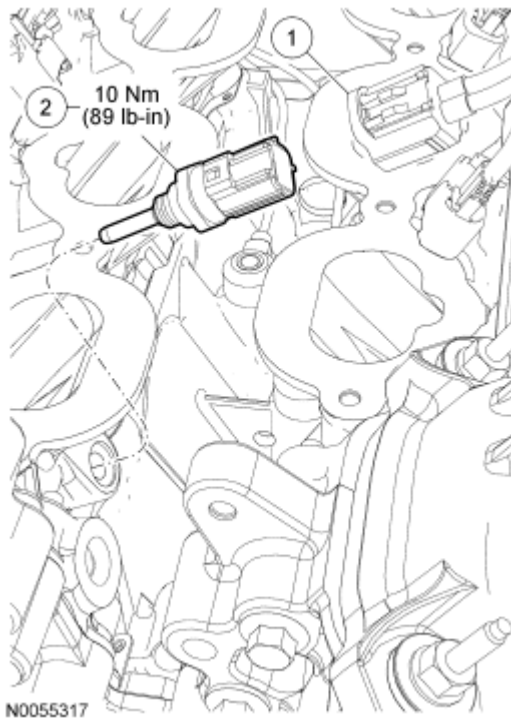


Fig. 5: Exploded View Of Cylinder Head Temperature (CHT) Sensor With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	14A464	Cylinder head temperature (CHT) sensor electrical connector
2	6G004	CHT sensor

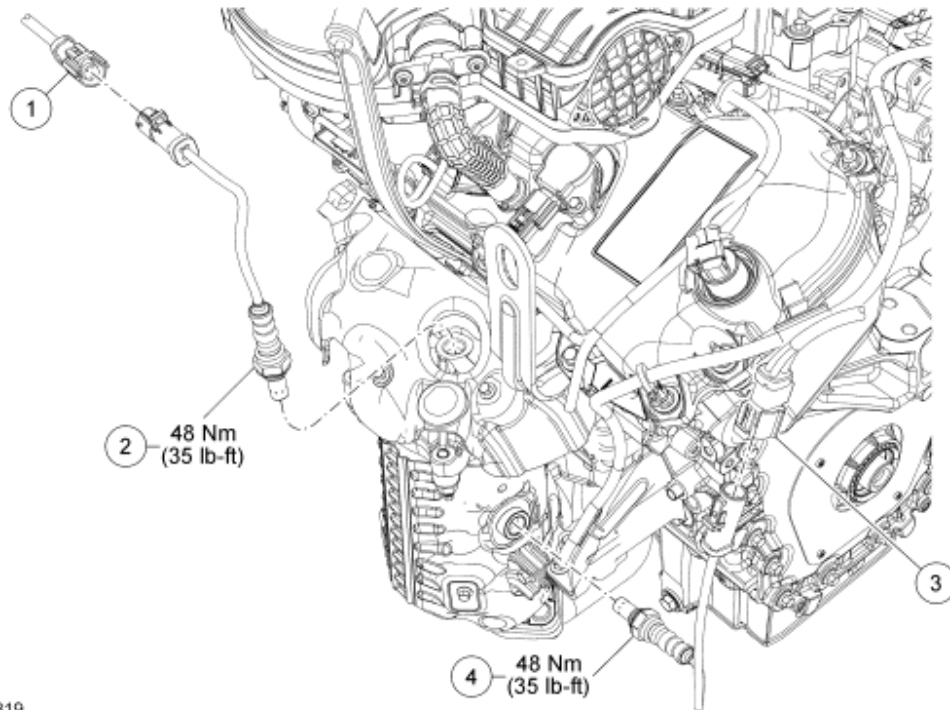
REMOVAL AND INSTALLATION

1. Remove the lower intake manifold. For additional information, refer to **ENGINE - 3.5L** article.
2. Disconnect the cylinder head temperature (CHT) sensor electrical connector.
3. Remove and discard the CHT sensor.
 - To install, tighten to 10 Nm (89 lb-in).
4. To install, reverse the removal procedure.
 - Do not reuse the CHT sensor, install a new sensor.

HEATED OXYGEN SENSOR (HO2S) AND CATALYST MONITOR SENSOR - EXPLODED VIEW

2008 Ford Edge SE

2008 ENGINE PERFORMANCE Electronic Engine Controls - Edge & MKX



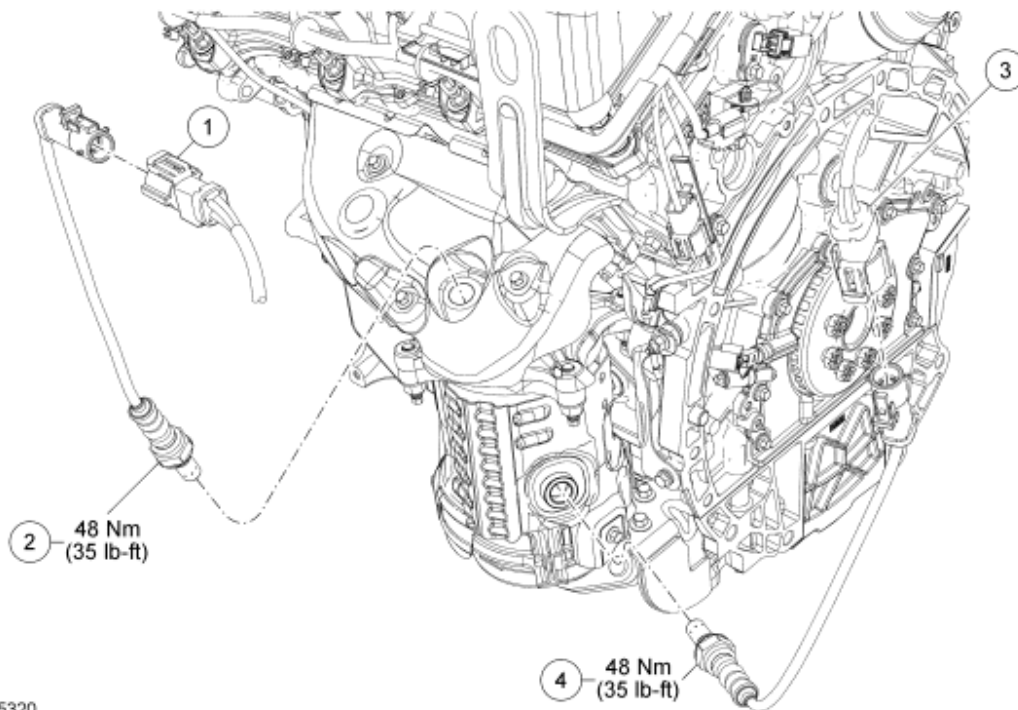
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Fig. 6: Exploded View Of Heated Oxygen Sensor (HO2S) & Catalyst Monitor Sensor With Torque Specifications - RH Sensors
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	14A464	Heated oxygen sensor (HO2S) electrical connector
2	9F472	HO2S
3	14A464	Catalyst monitor sensor electrical connector
4	9G444	Catalyst monitor sensor

2008 Ford Edge SE

2008 ENGINE PERFORMANCE Electronic Engine Controls - Edge & MKX



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Fig. 7: Exploded View Of Heated Oxygen Sensor (HO2S) & Catalyst Monitor Sensor With Torque Specifications - LH Sensors
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	14A464	Heated oxygen sensor (HO2S) electrical connector
2	9F472	HO2S
3	14A464	Catalyst monitor sensor electrical connector
4	9G444	Catalyst monitor sensor

1. For additional information, refer to the procedures in this article.

HEATED OXYGEN SENSOR (HO2S)

Special Tools

Illustration	Tool Name	Tool Number
 ST1447-A	Socket, Exhaust Gas Oxygen Sensor	303-476 (T94P-9472-A)

Material

Item	Specification
High Temperature Nickel Anti-Seize Lubricant XL-2 (US); CXG-2-B (Canada)	ESE-M12A4-A
Penetrating and Lock Lubricant XL-1 (US); CXC-51-A (Canada)	-

REMOVAL AND INSTALLATION

RH sensor

- 1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.

Both sensors

- 2. Disconnect the heated oxygen sensor (HO2S) electrical connector.

NOTE: **If necessary, lubricate the sensor threads with penetrating and lock lubricant to assist in removal.**

- 3. Using the special tool, remove the HO2S.
 - To install, tighten to 48 Nm (35 lb-ft).

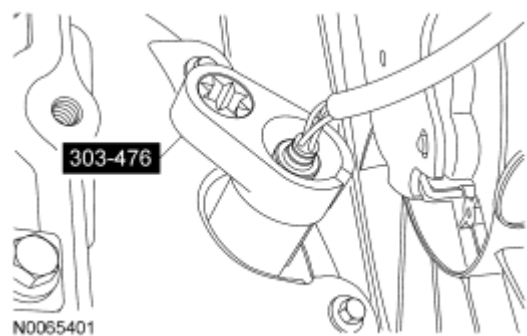


Fig. 8: Identifying Heated Oxygen Sensor (HO2S) & Special Tool (303-476)
Courtesy of FORD MOTOR CO.

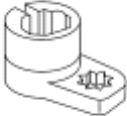
NOTE: **Apply a light coat of anti-seize lubricant to the threads of the HO2S.**

- 4. To install, reverse the removal procedure.

CATALYST MONITOR SENSOR

Special Tools

Illustration	Tool Name	Tool Number

 ST1447-A	Socket, Exhaust Gas Oxygen Sensor	303-476 (T94P-9472-A)
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Material

Item	Specification
High Temperature Nickel Anti-Seize Lubricant XL-2 (US); CXG-2-B (Canada)	ESE-M12A4-A
Penetrating and Lock Lubricant XL-1 (US); CXC-51-A (Canada)	-

REMOVAL AND INSTALLATION

- 1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.
- 2. Disconnect the catalyst monitor sensor electrical connector.

NOTE: **If necessary, lubricate the sensor threads with penetrating and lock lubricant to assist in removal.**

- 3. Using the special tool, remove the catalyst monitor sensor.
 - To install, tighten to 48 Nm (35 lb-ft).

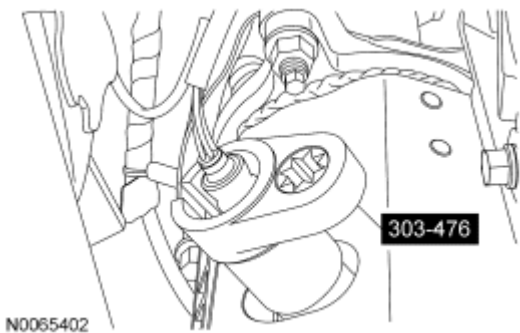


Fig. 9: Identifying Catalyst Monitor Sensor & Special Tool (303-476)
Courtesy of FORD MOTOR CO.

NOTE: **Apply a light coat of anti-seize lubricant to the threads of the catalyst monitor sensor.**

- 4. To install, reverse the removal procedure.

NOTE: LH shown, RH similar.

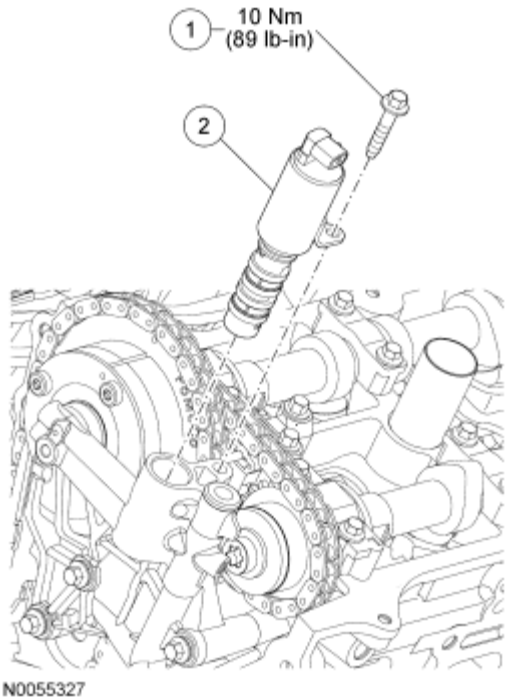


Fig. 10: Exploded View Of Variable Camshaft Timing (VCT) Oil Control Solenoid With Torque Specification

Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W500215	Variable camshaft timing (VCT) oil control solenoid bolt
2	6B297	VCT oil control solenoid

REMOVAL AND INSTALLATION

1. Remove the LH or RH valve cover. For additional information, refer to **ENGINE - 3.5L** article.
2. Remove the bolt and the variable camshaft timing (VCT) oil control solenoid.
 - To install, tighten to 10 Nm (89 lb-in).
3. To install, reverse the removal procedure.

KNOCK SENSOR (KS)

Material

Item	Specification
Motorcraft Premium Gold Engine Coolant with Bittering Agent (US only)	WSS-M97B51-A1

2008 Ford Edge SE

2008 ENGINE PERFORMANCE Electronic Engine Controls - Edge & MKX

VC-7-B (US); CVC-7-A (Canada); or equivalent
(yellow color)

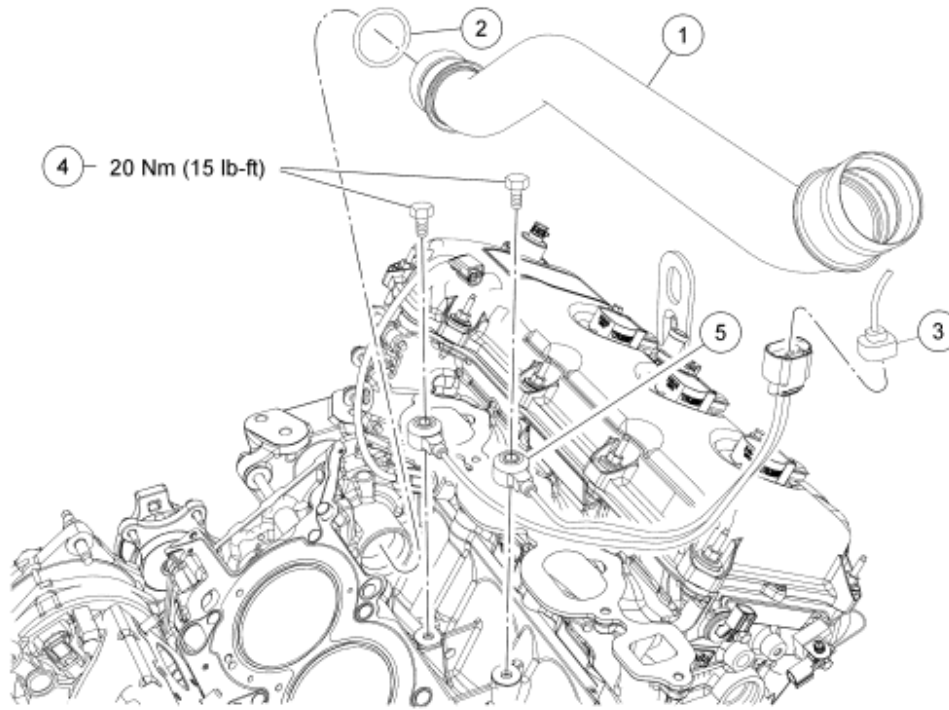


Fig. 11: Exploded View Of Knock Sensor (KS) With Torque Specification
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	9N271	Coolant tube
2	8565	O-ring seal
3	14A464	Knock sensor (KS) electrical connector
4	W500310	KS bolts (2 required)
5	12A699	KS

REMOVAL AND INSTALLATION

1. Remove the thermostat housing. For additional information, refer to **ENGINE COOLING** article.
2. Remove the lower intake manifold. For additional information, refer to **ENGINE - 3.5L** article.
3. Remove the coolant tube.
 - Discard the O-ring seal.
4. Disconnect the knock sensor (KS) electrical connector.
5. Remove the 2 bolts and the KS.
 - To install, tighten to 20 Nm (15 lb-ft).
6. To install, reverse the removal procedure.

2008 Ford Edge SE

2008 ENGINE PERFORMANCE Electronic Engine Controls - Edge & MKX

- Lubricate the new O-ring seal with clean engine coolant.

2008 Ford Edge SE

2008 ENGINE PERFORMANCE Electronic Engine Controls - Edge & MKX

2008 ENGINE PERFORMANCE

Electronic Engine Controls - Edge & MKX

SPECIFICATIONS

Material

Item	Specification	Fill Capacity
High Temperature Nickel Anti-Seize Lubricant XL-2 (US); CXG-2-B (Canada)	ESE-M12A4-A	-
Motorcraft Premium Gold Engine Coolant with Bittering Agent (US only) VC-7-B (US); CVC-7-A (Canada); or equivalent (yellow color)	WSS-M97B51-A1	-
Motorcraft SAE 5W-20 Premium Synthetic Blend Motor Oil XO-5W20-QSP (US); Motorcraft SAE 5W-20 Super Premium Motor Oil CXO-5W20-LSP12 (Canada); or equivalent	WSS-M2C930-A	-
Penetrating and Lock Lubricant XL-1 (US); CXC-51-A (Canada)	-	-

TORQUE SPECIFICATIONS

Description	Nm	lb-ft	lb-in
Camshaft position (CMP) sensor bolt	10	-	89
Catalyst monitor sensor	48	35	-
Crankshaft position (CKP) sensor bolt	10	-	89
Cylinder head temperature (CHT) sensor	10	-	89
Heat shield bolt and nut	10	-	89
Heated oxygen sensor (HO2S)	48	35	-
Knock sensor (KS) bolt	20	15	-
Mass airflow (MAF) sensor screws	2	-	18
PCM nuts	5	-	44
Variable camshaft timing (VCT) oil control solenoid bolt	10	-	89

DESCRIPTION AND OPERATION

ELECTRONIC ENGINE CONTROLS

The electronic engine controls consist of the:

- PCM.
- cylinder head temperature (CHT) sensor.

2008 Ford Edge SE

2008 ENGINE PERFORMANCE Electronic Engine Controls - Edge & MKX

- camshaft position (CMP) sensor.
- crankshaft position (CKP) sensor.
- mass air flow (MAF) sensor.
- heated oxygen sensor (HO2S).
- catalyst monitor sensor.
- variable camshaft timing (VCT) oil control solenoid.
- knock sensor (KS).

The PCM carries out the following functions:

- accepts input from various engine sensors to compute the fuel flow rate necessary to maintain a prescribed air/fuel ratio throughout the entire engine operational range.
- outputs a command to the fuel injectors to meter the appropriate quantity of fuel.

The CHT sensor:

- sends the PCM a signal indicating cylinder head temperature.
- resistance decreases as cylinder head temperature increases.

The CMP sensor:

- sends the PCM a signal indicating camshaft position used for fuel synchronization.

The CKP sensor:

- sends the PCM a signal indicating crankshaft position.
- is essential for calculating spark timing.

The MAF sensor:

- uses a hot wire sensing element to measure the amount of air entering the engine. Air passing over the hot wire causes it to cool.

The HO2S:

- creates a voltage signal dependent on exhaust oxygen content.
- provides feedback information to the PCM used to calculate fuel delivery.

The catalyst monitor sensor:

- monitors oxygen content after it flows through the catalytic converter.
- provides a voltage to the PCM used to calculate catalytic converter integrity.

The KS:

- is used to detect engine detonation.
- sends a voltage signal to the PCM.
- is able to provide a signal which retards the ignition timing, as necessary.

The VCT oil control solenoid:

- is an electrically controlled hydraulic valve that directs engine oil to the camshaft phaser. Once the PCM transmits a signal, the solenoid moves a valve spool, directing oil into the camshaft phaser cavity. This action changes valve timing by either inducing an advance or retard condition. The camshaft is, thereby repositioned in relation to crankshaft timing and allows for optimum engine performance and lower emissions.

DIAGNOSTIC TESTS

ELECTRONIC ENGINE CONTROLS

Refer to the [Introduction - Gasoline Engines](#) article.

REMOVAL AND INSTALLATION

CAMSHAFT POSITION (CMP) SENSOR

Material

Item	Specification
Motorcraft SAE 5W-20 Premium Synthetic Blend Motor Oil XO-5W20-QSP (US); Motorcraft SAE 5W-20 Super Premium Motor Oil CXO-5W20-LSP12 (Canada); or equivalent	WSS-M2C930-A

2008 Ford Edge SE

2008 ENGINE PERFORMANCE Electronic Engine Controls - Edge & MKX

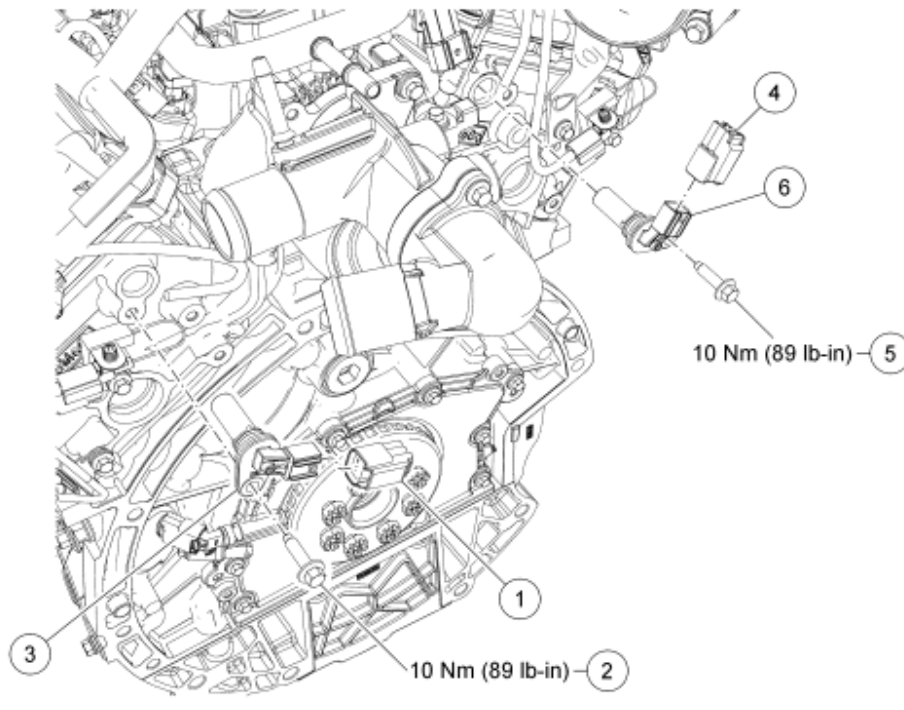


Fig. 1: Exploded View Of Camshaft Position (CMP) Sensor With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	14A464	LH camshaft position sensor (CMP) electrical connector
2	W500214	LH CMP bolt
3	6B288	LH CMP
4	14A464	RH CMP electrical connector
5	W500214	RH CMP bolt
6	6B288	RH CMP

REMOVAL AND INSTALLATION

RH sensor

1. Remove the air cleaner outlet pipe. For additional information, refer to **INTAKE AIR DISTRIBUTION AND FILTERING** article.

LH sensor

2. Remove the air cleaner assembly. For additional information, refer to **INTAKE AIR DISTRIBUTION AND FILTERING** article.

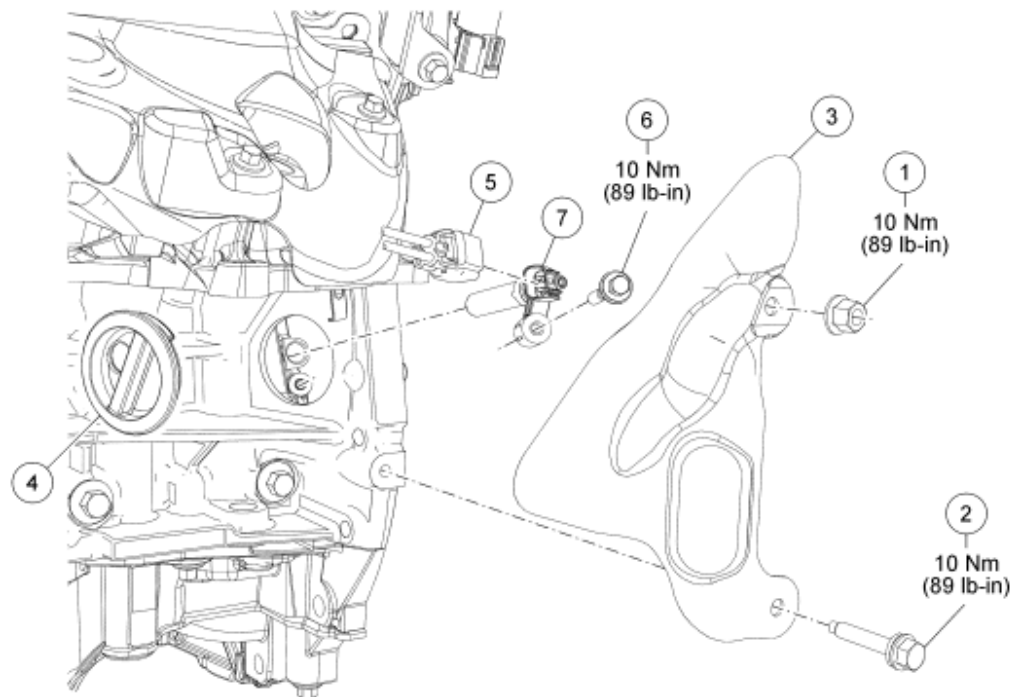
Both sensors

3. Disconnect the camshaft position sensor (CMP) electrical connector.
4. Remove the bolt and the CMP sensor.
 - To install, tighten to 10 Nm (89 lb-in).

NOTE: Lubricate the CMP O-ring seal with clean engine oil.

5. To install, reverse the removal procedure.

CRANKSHAFT POSITION (CKP) SENSOR



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Fig. 2: Exploded View Of Crankshaft Position (CKP) Sensor With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W701542	Heat shield nut
2	W701669	Heat shield bolt
3	6K342	Heat shield
4	6C070	Rubber grommet cover
5	14A464	Crankshaft position (CKP) sensor electrical connector
6	6D327	CKP sensor bolt
7	6C315	CKP sensor

REMOVAL AND INSTALLATION

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.
2. Remove the LH catalytic converter. For additional information, refer to **EXHAUST SYSTEM** article.
3. Remove the bolt, nut and the heat shield.
 - To install, tighten to 10 Nm (89 lb-in).
4. Remove the rubber grommet cover.
5. Disconnect the crankshaft position (CKP) sensor electrical connector.
6. Remove the bolt and the CKP sensor.
 - To install, tighten to 10 Nm (89 lb-in).
7. To install, reverse the removal procedure.

POWERTRAIN CONTROL MODULE (PCM)

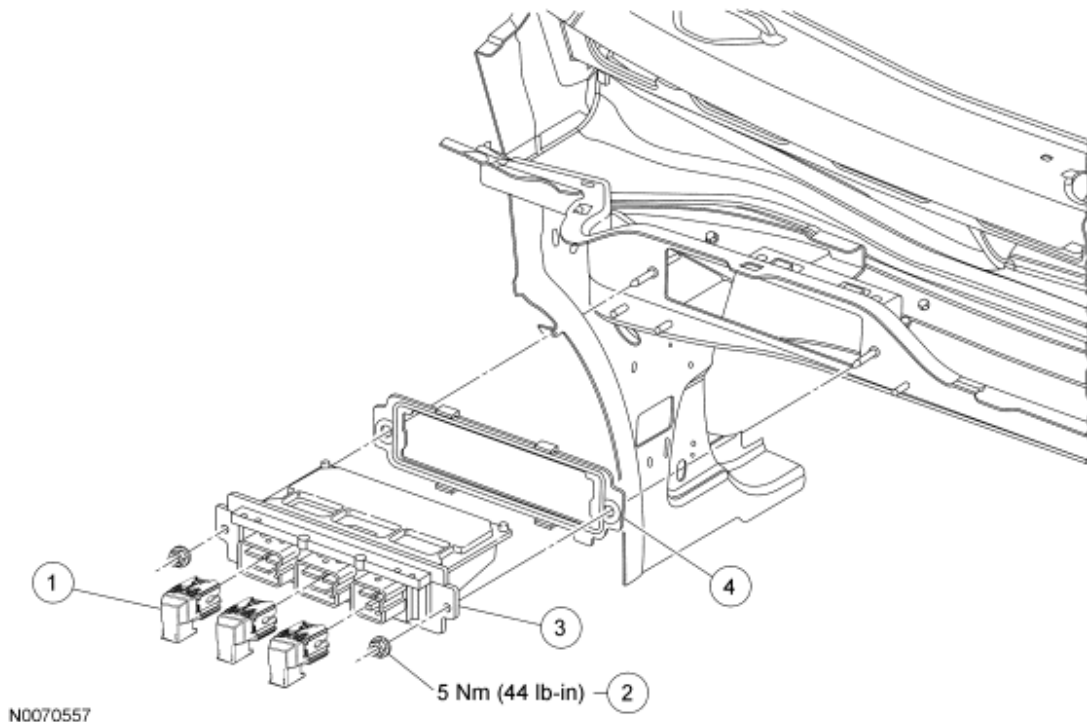


Fig. 3: Exploded View Of Powertrain Control Module (PCM) With Torque Specification
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	14A464	PCM electrical connector (part of 12B637) (3 required)
2	N804795	PCM nut (2 required)
3	12A650	PCM
4	12A696	PCM gasket

REMOVAL

NOTE: PCM replacement **DOES NOT** require new keys or programming of keys.

1. Retrieve the module configuration. Carry out the module configuration retrieval steps of the Programmable Module Installation procedure. For additional information, refer to **MODULE CONFIGURATION** article.
2. Disconnect the 3 PCM electrical connectors.
3. Remove the 2 nuts and the PCM.
 - Remove the gasket.

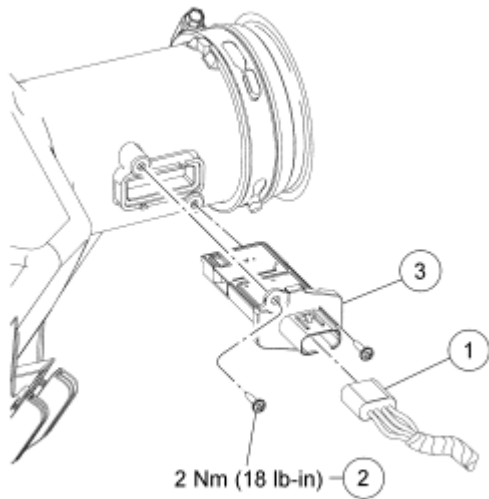
INSTALLATION

1. Install the gasket and the PCM.
2. Install the 2 PCM nuts.
 - Tighten to 5 Nm (44 lb-in).
3. Connect the 3 PCM electrical connectors.
4. Restore the module configuration. Carry out the module configuration restore steps of the Programmable Module Installation procedure. For additional information, refer to **MODULE CONFIGURATION** article.
5. Reprogram the passive anti-theft system (PATS). Carry out the Parameter Reset procedure. For additional information, refer to **ANTI-THEFT - PASSIVE ANTI-THEFT SYSTEM (PATS)** article.

MASS AIR FLOW (MAF) SENSOR

2008 Ford Edge SE

2008 ENGINE PERFORMANCE Electronic Engine Controls - Edge & MKX



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Fig. 4: Exploded View Of Mass Air Flow (MAF) Sensor With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	14A464	Mass air flow (MAF) sensor electrical connector
2	W709287	MAF sensor screw (2 required)
3	12B579	MAF sensor

REMOVAL AND INSTALLATION

1. Disconnect the mass air flow (MAF) sensor electrical connector.
2. Remove the 2 screws and the MAF sensor.
 - To install, tighten to 2 Nm (18 lb-in).
3. To install, reverse the removal procedure.

CYLINDER HEAD TEMPERATURE (CHT) SENSOR

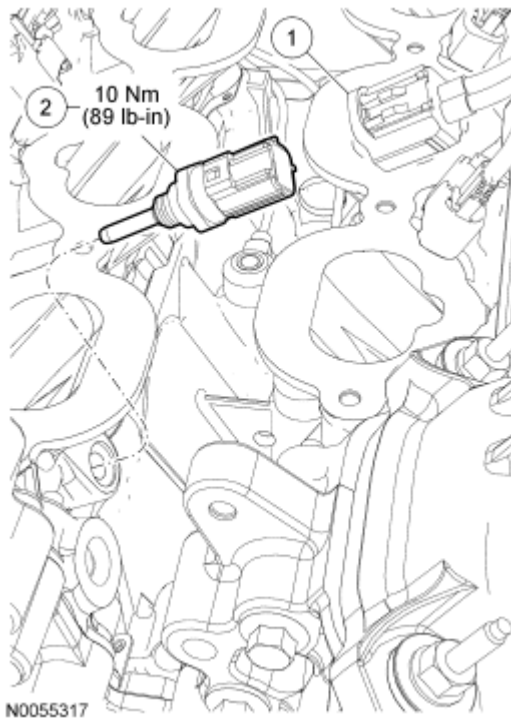


Fig. 5: Exploded View Of Cylinder Head Temperature (CHT) Sensor With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	14A464	Cylinder head temperature (CHT) sensor electrical connector
2	6G004	CHT sensor

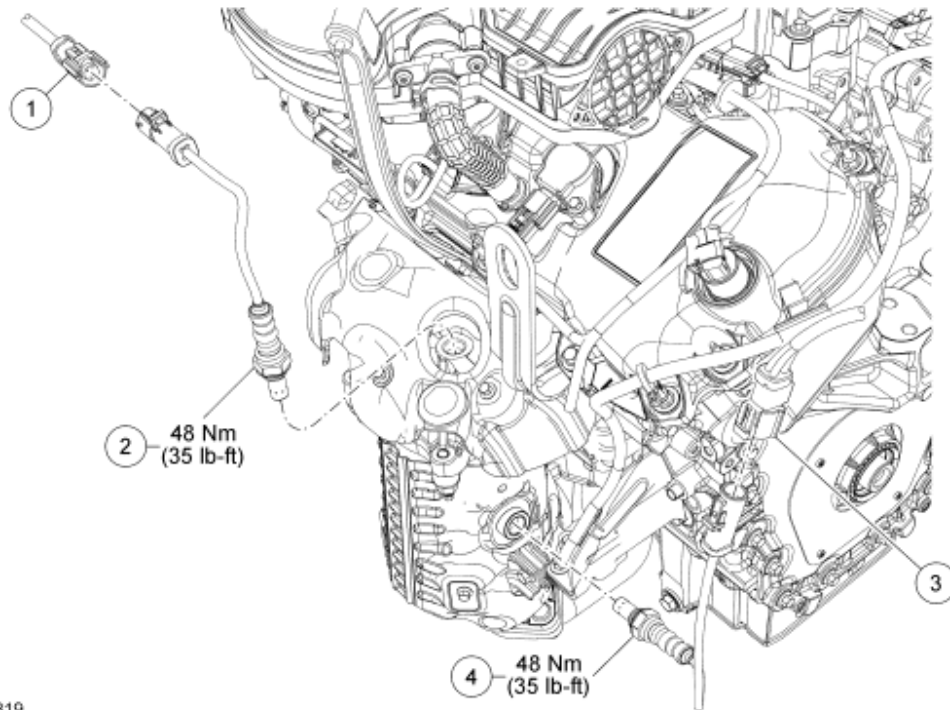
REMOVAL AND INSTALLATION

1. Remove the lower intake manifold. For additional information, refer to **ENGINE - 3.5L** article.
2. Disconnect the cylinder head temperature (CHT) sensor electrical connector.
3. Remove and discard the CHT sensor.
 - To install, tighten to 10 Nm (89 lb-in).
4. To install, reverse the removal procedure.
 - Do not reuse the CHT sensor, install a new sensor.

HEATED OXYGEN SENSOR (HO2S) AND CATALYST MONITOR SENSOR - EXPLODED VIEW

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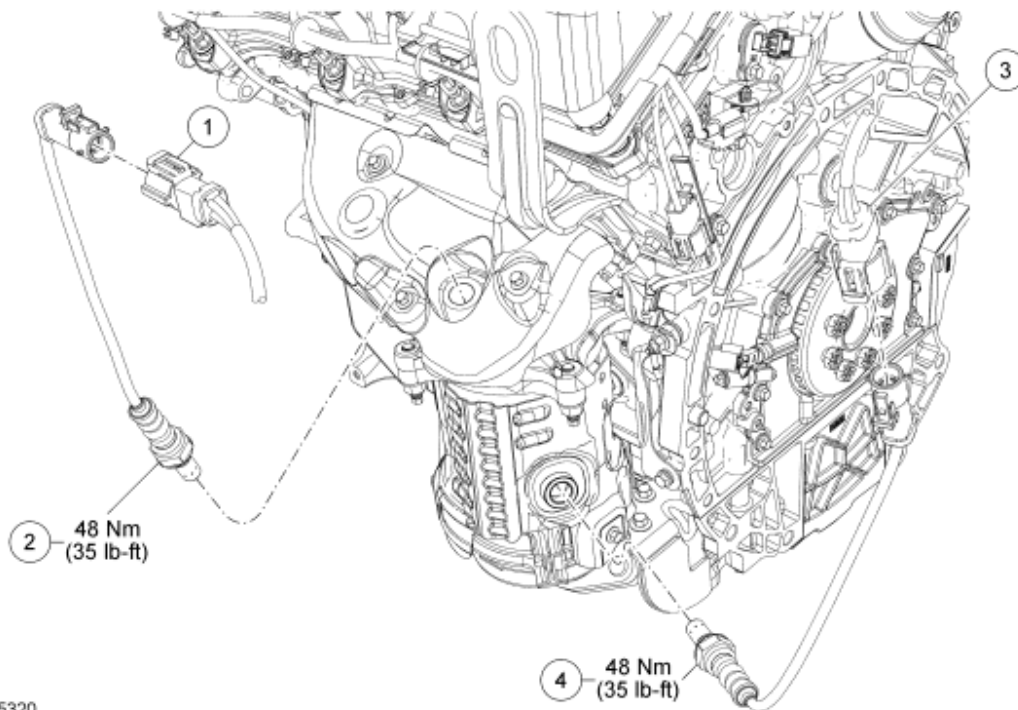
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Fig. 6: Exploded View Of Heated Oxygen Sensor (HO2S) & Catalyst Monitor Sensor With Torque Specifications - RH Sensors
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	14A464	Heated oxygen sensor (HO2S) electrical connector
2	9F472	HO2S
3	14A464	Catalyst monitor sensor electrical connector
4	9G444	Catalyst monitor sensor

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Fig. 7: Exploded View Of Heated Oxygen Sensor (HO2S) & Catalyst Monitor Sensor With Torque Specifications - LH Sensors
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	14A464	Heated oxygen sensor (HO2S) electrical connector
2	9F472	HO2S
3	14A464	Catalyst monitor sensor electrical connector
4	9G444	Catalyst monitor sensor

1. For additional information, refer to the procedures in this article.

HEATED OXYGEN SENSOR (HO2S)

Special Tools

Illustration	Tool Name	Tool Number
 ST1447-A	Socket, Exhaust Gas Oxygen Sensor	303-476 (T94P-9472-A)

Material

Item	Specification
High Temperature Nickel Anti-Seize Lubricant XL-2 (US); CXG-2-B (Canada)	ESE-M12A4-A
Penetrating and Lock Lubricant XL-1 (US); CXC-51-A (Canada)	-

REMOVAL AND INSTALLATION

RH sensor

- 1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.

Both sensors

- 2. Disconnect the heated oxygen sensor (HO2S) electrical connector.

NOTE: **If necessary, lubricate the sensor threads with penetrating and lock lubricant to assist in removal.**

- 3. Using the special tool, remove the HO2S.
 - To install, tighten to 48 Nm (35 lb-ft).

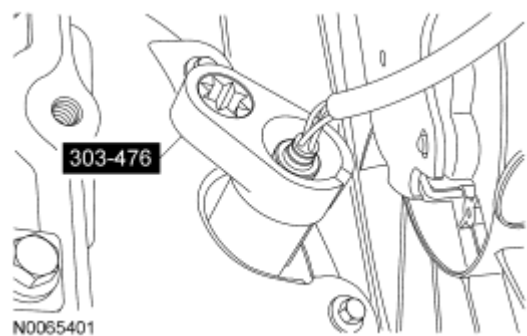


Fig. 8: Identifying Heated Oxygen Sensor (HO2S) & Special Tool (303-476)
Courtesy of FORD MOTOR CO.

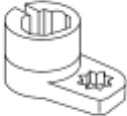
NOTE: **Apply a light coat of anti-seize lubricant to the threads of the HO2S.**

- 4. To install, reverse the removal procedure.

CATALYST MONITOR SENSOR

Special Tools

Illustration	Tool Name	Tool Number

 ST1447-A	Socket, Exhaust Gas Oxygen Sensor	303-476 (T94P-9472-A)
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Material

Item	Specification
High Temperature Nickel Anti-Seize Lubricant XL-2 (US); CXG-2-B (Canada)	ESE-M12A4-A
Penetrating and Lock Lubricant XL-1 (US); CXC-51-A (Canada)	-

REMOVAL AND INSTALLATION

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.
2. Disconnect the catalyst monitor sensor electrical connector.

NOTE: **If necessary, lubricate the sensor threads with penetrating and lock lubricant to assist in removal.**

3. Using the special tool, remove the catalyst monitor sensor.
 - To install, tighten to 48 Nm (35 lb-ft).

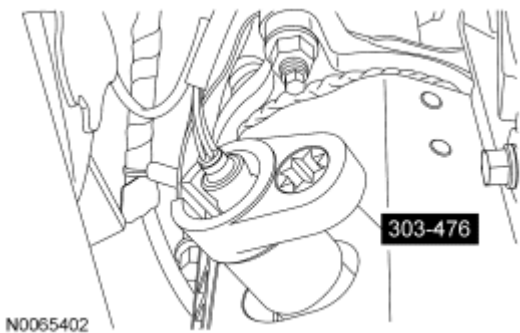


Fig. 9: Identifying Catalyst Monitor Sensor & Special Tool (303-476)
Courtesy of FORD MOTOR CO.

NOTE: **Apply a light coat of anti-seize lubricant to the threads of the catalyst monitor sensor.**

4. To install, reverse the removal procedure.

NOTE: LH shown, RH similar.

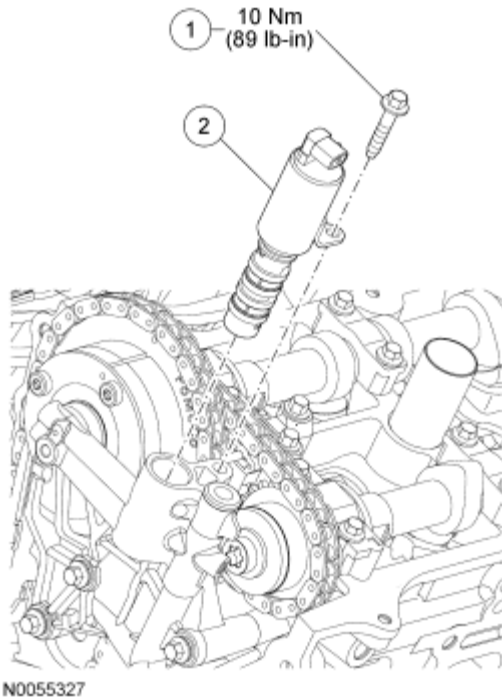


Fig. 10: Exploded View Of Variable Camshaft Timing (VCT) Oil Control Solenoid With Torque Specification

Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W500215	Variable camshaft timing (VCT) oil control solenoid bolt
2	6B297	VCT oil control solenoid

REMOVAL AND INSTALLATION

1. Remove the LH or RH valve cover. For additional information, refer to **ENGINE - 3.5L** article.
2. Remove the bolt and the variable camshaft timing (VCT) oil control solenoid.
 - To install, tighten to 10 Nm (89 lb-in).
3. To install, reverse the removal procedure.

KNOCK SENSOR (KS)

Material

Item	Specification
Motorcraft Premium Gold Engine Coolant with Bittering Agent (US only)	WSS-M97B51-A1

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VC-7-B (US); CVC-7-A (Canada); or equivalent
(yellow color)

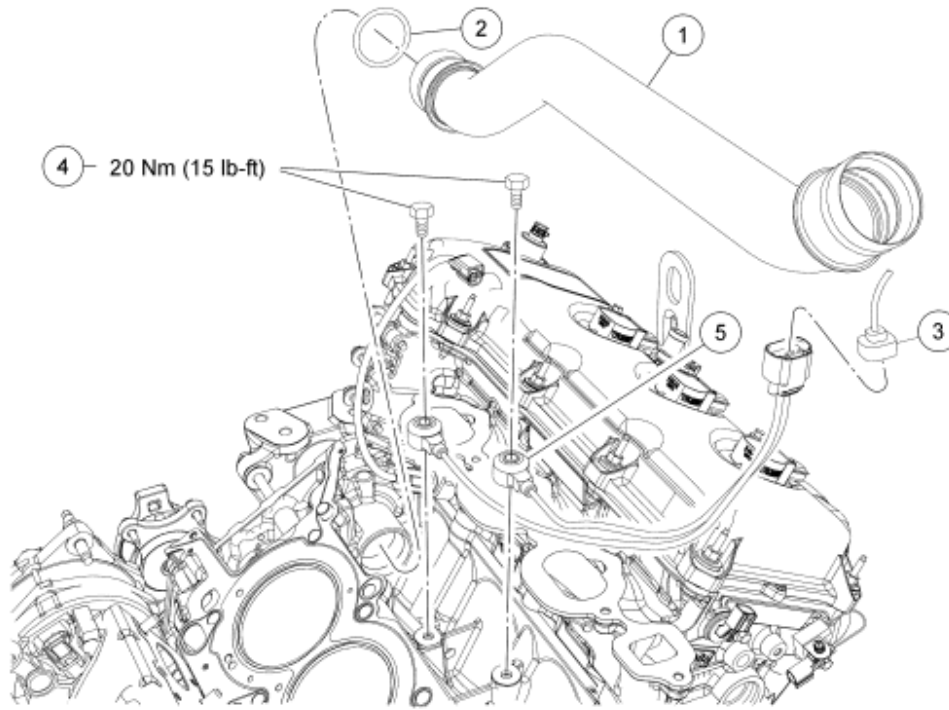


Fig. 11: Exploded View Of Knock Sensor (KS) With Torque Specification
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	9N271	Coolant tube
2	8565	O-ring seal
3	14A464	Knock sensor (KS) electrical connector
4	W500310	KS bolts (2 required)
5	12A699	KS

REMOVAL AND INSTALLATION

1. Remove the thermostat housing. For additional information, refer to **ENGINE COOLING** article.
2. Remove the lower intake manifold. For additional information, refer to **ENGINE - 3.5L** article.
3. Remove the coolant tube.
 - Discard the O-ring seal.
4. Disconnect the knock sensor (KS) electrical connector.
5. Remove the 2 bolts and the KS.
 - To install, tighten to 20 Nm (15 lb-ft).
6. To install, reverse the removal procedure.

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- Lubricate the new O-ring seal with clean engine coolant.

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SPECIFICATIONS

Material

Item	Specification	Fill Capacity
Motorcraft SAE 5W-30 Premium Synthetic Blend Motor Oil XO-5W30-QSP (US); Motorcraft SAE 5W-30 Super Premium Motor Oil CXO-5W30-LSP12 (Canada); or equivalent	WSS-M2C929-A	-

TORQUE SPECIFICATIONS

Description	Nm	lb-in
PCV fitting screws	6	53

DESCRIPTION AND OPERATION

ENGINE EMISSION CONTROL

NOTE: Do not permanently remove or render inoperative any part of the vehicle emission control system including related hardware. Failure to comply may violate applicable state and federal laws.

The engine emission control system consists of the PCV system.

The PCV system consists of the:

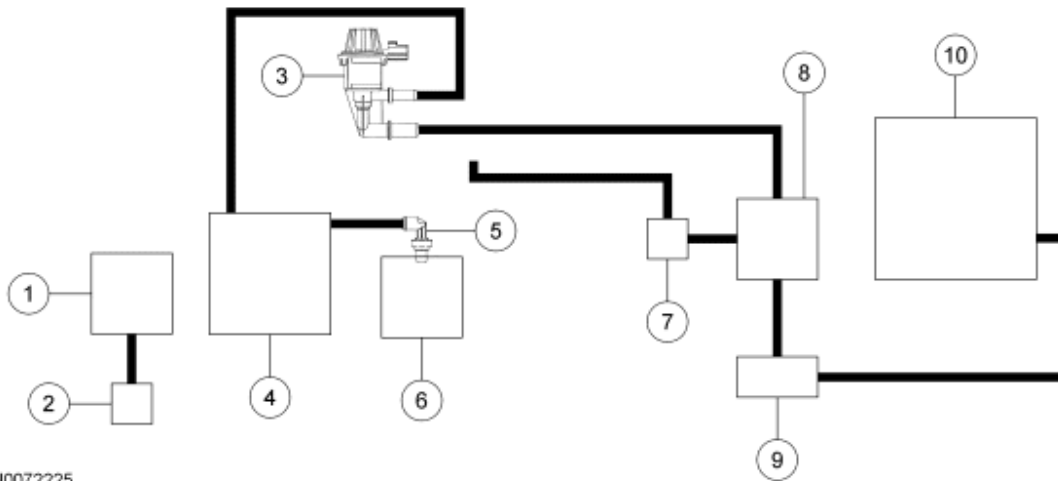
- crankcase ventilation tube.
- PCV fitting.
- PCV valve.

The PCV system uses intake manifold vacuum to ventilate blow-by fumes from the crankcase and return the vapors to the intake manifold for combustion. The PCV valve varies the amount of blow-by gases returned to the intake manifold based on available engine vacuum. The PCV valve also prevents the entry of combustion backfiring into the crankcase.

Vehicle Emission Vacuum Routing

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Fig. 1: Identifying Vehicle Emission Vacuum Routing
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	6A505	LH valve cover
2	9F805	Air cleaner outlet pipe
3	9G641	Evaporative emission (EVAP) canister purge valve
4	9S455	Upper intake manifold
5	6A666	PCV valve
6	6582	RH valve cover
7	9G676	EVAP canister vent solenoid and dust separator assembly
8	9E857	EVAP canister
9	-	Fuel tank pressure (FTP) sensor (part of 9S284)
10	9K007	Fuel tank

DIAGNOSTIC TESTS

ENGINE EMISSION CONTROL

Refer to the [Introduction - Gasoline Engines](#) article.

REMOVAL AND INSTALLATION

POSITIVE CRANKCASE VENTILATION (PCV) SYSTEM COMPONENTS

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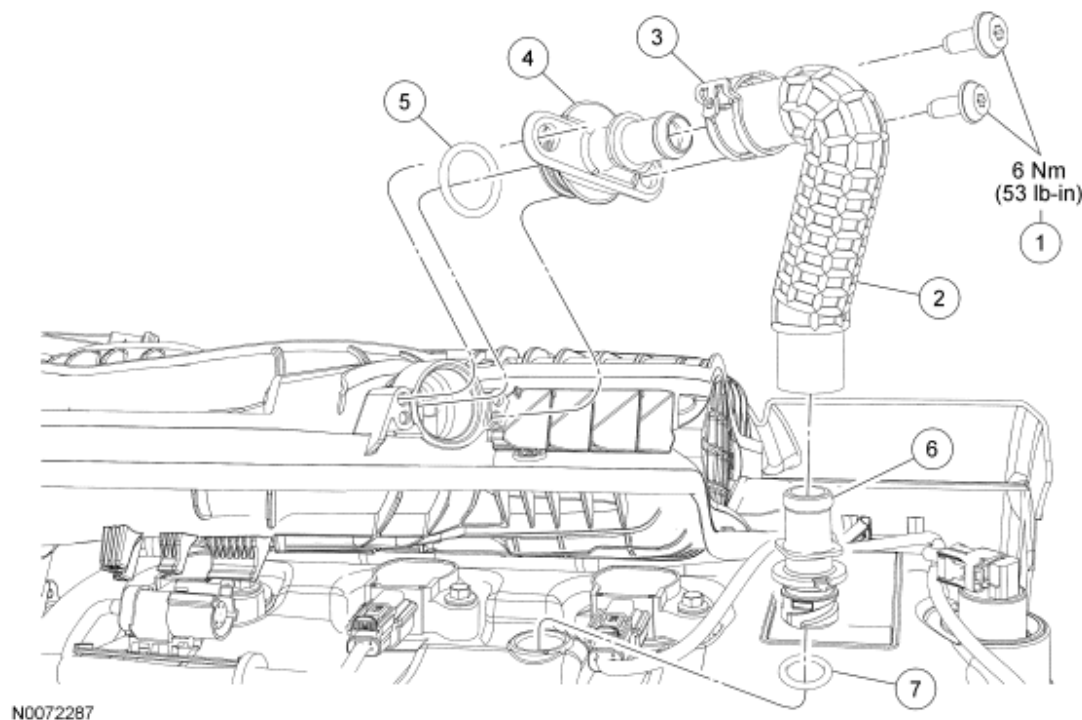


Fig. 2: Identifying Positive Crankcase Ventilation (PCV) System Components
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	PCV fitting screws (2 required)
2	6A664	Crankcase ventilation tube
3	-	Crankcase ventilation tube spring clamp (part of 6K817)
4	9F695	PCV fitting
5	9J469	PCV fitting O-ring seal
6	6A666	PCV valve
7	-	PCV valve O-ring seal (part of 6A666)

1. For additional information, refer to the procedures in this article.

POSITIVE CRANKCASE VENTILATION (PCV) VALVE

Material

Item	Specification
Motorcraft SAE 5W-20 Premium Synthetic Blend Motor Oil XO-5W20-QSP (US); Motorcraft SAE 5W-20 Super Premium Motor Oil CXO-5W20-LSP12 (Canada); or equivalent	WSS-M2C930-A

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REMOVAL AND INSTALLATION

1. Disconnect the crankcase ventilation tube from the PCV valve.

CAUTION: A new positive crankcase ventilation (PCV) valve must be installed if removed from the valve cover. During removal, damage will occur to the locking mechanism on the PCV valve.

NOTE: To install, apply clean engine oil to the O-ring seal.

2. Rotate the PCV valve counterclockwise and remove from the valve cover.
 - Discard the PCV valve.
3. To install, reverse the removal procedure.